Tutorial 5: Local Density and Clustering

Three required tasks

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How to use this tutorial

- Attempt each problem before reading its solution.
- Write the main calculation or argument clearly.
- Explain what the result means, not only the final number or formula.
- Complete Tasks 1–3 first. The final task is an optional challenge.

Task 1: KDE and bandwidth selection

Consider the observations:

$$x_1 = 0, \qquad x_2 = 2, \qquad x_3 = 3.$$

Use the triangular kernel:

$$K(u) = \begin{cases} 1 - |u|, & |u| \le 1, \\ 0, & |u| > 1. \end{cases}$$

The KDE is:

$$\hat{p}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right).$$

1. Compute $\hat{p}_h(2)$ for $h = 1$ and $h = 2$.
2. Compute $\hat{p}_h(1)$ for $h = 1$ and $h = 2$.
3. A validation observation is $x^{\text{val}} = 1$. Which bandwidth assigns it the larger density?
4. Based on these calculations, describe what changing h changes in the KDE.

Solution 1: density at $x = 2$

1. For $h = 1$:

$$\hat{p}_1(2) = \frac{1}{3}\left[K(2) + K(0) + K(-1)\right].$$

Using the triangular kernel:

$$K(2) = 0, \qquad K(0) = 1, \qquad K(-1) = 0.$$

Therefore:

$$\boxed{\hat{p}_1(2) = \frac{1}{3}}.$$

For $h = 2$:

$$\hat{p}_2(2) = \frac{1}{6}\left[K(1) + K(0) + K(-0.5)\right].$$

Since:

$$K(1) = 0, \qquad K(0) = 1, \qquad K(-0.5) = 0.5,$$

we obtain:

$$\boxed{\hat{p}_2(2) = \frac{1}{4}}.$$

Solution 1: density at $x = 1$

2. For $h = 1$:

$$\hat{p}_1(1) = \frac{1}{3}\left[K(1) + K(-1) + K(-2)\right].$$

All three kernel values are zero:

$$K(1) = K(-1) = K(-2) = 0.$$

Hence:

$$\boxed{\hat{p}_1(1) = 0}.$$

For $h = 2$:

$$\hat{p}_2(1) = \frac{1}{6}\left[K(0.5) + K(-0.5) + K(-1)\right].$$

Thus:

$$\hat{p}_2(1) = \frac{1}{6}\left[0.5 + 0.5 + 0\right] = \boxed{\frac{1}{6}}.$$

Solution 1: bandwidth selection

3. Compare the validation densities.
For $x^{\text{val}} = 1$:

$$\hat{p}_1(1) = 0, \qquad \hat{p}_2(1) = \frac{1}{6}.$$

Therefore:

$$\boxed{h = 2}$$

assigns the larger density to the validation observation.
4. Effect of changing the bandwidth.
Increasing h makes each kernel contribution wider.
In this example: $h = 1$ leaves a zero-density gap at $x = 1$, while: $h = 2$ allows contributions from nearby observations to overlap.
Thus a larger bandwidth produces more smoothing and shares information over a larger neighborhood.

Task 2: Mean-shift iterations

Consider:

$$x_1 = 0, \qquad x_2 = 1, \qquad x_3 = 4, \qquad x_4 = 5.$$

Use Gaussian mean-shift weights with $h = 1$:

$$w_i(x) = \exp\left(-\frac{(x - x_i)^2}{2}\right).$$

Start from:

$$x^{(0)} = 2.$$

1. Compute the four weights $w_i(x^{(0)})$.
2. Compute the local weighted mean $m(x^{(0)})$.
3. Compute the mean-shift vector $s(x^{(0)})$ and obtain $x^{(1)}$.
4. Recompute the weights at $x^{(1)}$ and obtain $x^{(2)}$.
5. From the two updates, identify which group the trajectory is moving toward.

You may use a calculator and report numerical values to three decimal places.

Solution 2: first set of weights

1. At $x^{(0)} = 2$:

$$w_i = \exp\left(-\frac{(2 - x_i)^2}{2}\right).$$

Therefore:

$$w_1 = e^{-2} \approx 0.135, \qquad w_2 = e^{-1/2} \approx 0.607,$$
$$w_3 = e^{-2} \approx 0.135, \qquad w_4 = e^{-9/2} \approx 0.011.$$

Hence:

$$\boxed{(w_1, w_2, w_3, w_4) \approx (0.135, 0.607, 0.135, 0.011)}.$$

The point at $x_2 = 1$ receives the largest weight because it is closest to the current location.

Solution 2: first mean-shift update

2. Compute the weighted mean.

$$m(x^{(0)}) = \frac{0.135(0) + 0.607(1) + 0.135(4) + 0.011(5)}{0.135 + 0.607 + 0.135 + 0.011}.$$

Thus:

$$m(x^{(0)}) \approx \frac{1.203}{0.888} \approx \boxed{1.355}.$$

3. Compute the mean-shift vector.

$$s(x^{(0)}) = m(x^{(0)}) - x^{(0)} = 1.355 - 2.$$

Therefore:

$$\boxed{s(x^{(0)}) \approx -0.645}, \qquad \boxed{x^{(1)} \approx 1.355}.$$

The first update moves to the left.

Solution 2: recompute the weights

4. At $x^{(1)} \approx 1.355$:

$$w_i^{(1)} = \exp\left(-\frac{(1.355 - x_i)^2}{2}\right).$$

This gives approximately:

$$w_1^{(1)} \approx 0.399, \qquad w_2^{(1)} \approx 0.939,$$
$$w_3^{(1)} \approx 0.030, \qquad w_4^{(1)} \approx 0.001.$$

Therefore:

$$\boxed{(w_1^{(1)}, w_2^{(1)}, w_3^{(1)}, w_4^{(1)}) \approx (0.399, 0.939, 0.030, 0.001)}.$$

The observations near 0 and 1 now dominate the local weighted mean.

Solution 2: second update

Using the new weights:

$$x^{(2)} = \frac{0.399(0) + 0.939(1) + 0.030(4) + 0.001(5)}{0.399 + 0.939 + 0.030 + 0.001}.$$

Hence:

$$x^{(2)} \approx \frac{1.064}{1.369} \approx \boxed{0.778}.$$

5. Direction of the trajectory.
The first two updates are:

$$2 \to 1.355 \to 0.778.$$

Therefore the trajectory is moving toward the left high-density group near:

$$\boxed{\{0, 1\}}.$$

Repeated updates would continue recomputing the local weights until the location stabilizes near a density mode.

Task 3: Graph cuts and the Laplacian

Consider:

$$\boldsymbol{W} = \begin{bmatrix} 0 & 3 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 3 & 0 \end{bmatrix}.$$

Compare:

$$\mathcal{P}_1 : \{1, 2\} \mid \{3, 4\}, \qquad \mathcal{P}_2 : \{1\} \mid \{2, 3, 4\}.$$

1. Compute the degree matrix $\boldsymbol{D}$ and Laplacian $\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{W}$.
2. Compute $\mathrm{cut}(A, B)$ for both partitions.
3. Compute $\mathrm{Ncut}(A, B)$ for both partitions.
4. For $\boldsymbol{f} = [1, 1, 0, 0]^\top$, compute $\boldsymbol{f}^\top \boldsymbol{L} \boldsymbol{f}$.
5. Determine which partition is preferred by normalized cut.

Solution 3: degree matrix

1. Compute the weighted degrees.
The row sums of $\boldsymbol{W}$ are:

$$d_1 = 3, \qquad d_2 = 4, \qquad d_3 = 4, \qquad d_4 = 3.$$

Therefore:

$$\boxed{\boldsymbol{D} = \mathrm{diag}(3, 4, 4, 3)}.$$

The degree matrix records the total similarity connected to each node.

Solution 3: graph Laplacian

Using:

$$\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{W},$$

we obtain:

$$\boxed{\boldsymbol{L} = \begin{bmatrix} 3 & -3 & 0 & 0 \\ -3 & 4 & -1 & 0 \\ 0 & -1 & 4 & -3 \\ 0 & 0 & -3 & 3 \end{bmatrix}}.$$

The diagonal entries contain total connection strength, while the negative off-diagonal entries represent pairwise connections.
Each row sums to zero:

$$\boldsymbol{L}\mathbf{1} = \mathbf{0}.$$

Solution 3: cut values

2. Partition $\mathcal{P}_1$:

$$\{1, 2\} \mid \{3, 4\}.$$

Only the edge between nodes 2 and 3 crosses the partition, with weight 1:

$$\boxed{\text{cut}(\mathcal{P}_1) = 1}.$$

For:

$$\mathcal{P}_2 : \{1\} \mid \{2, 3, 4\},$$

the edge between nodes 1 and 2 crosses the partition, with weight 3:

$$\boxed{\text{cut}(\mathcal{P}_2) = 3}.$$

Thus $\mathcal{P}_1$ cuts a much weaker connection.

Solution 3: normalized cut for $\mathcal{P}_1$

For:

$$\mathcal{P}_1 = \{1, 2\} \mid \{3, 4\},$$

the graph volumes are:

$$\mathrm{vol}(\{1, 2\}) = 3 + 4 = 7,$$
$$\mathrm{vol}(\{3, 4\}) = 4 + 3 = 7.$$

Therefore:

$$\mathrm{Ncut}(\mathcal{P}_1) = \frac{1}{7} + \frac{1}{7} = \boxed{\frac{2}{7}} \approx 0.286.$$

Solution 3: normalized cut for $\mathcal{P}_2$

For:

$$\mathcal{P}_2 = \{1\} \mid \{2, 3, 4\},$$

the graph volumes are:

$$\mathrm{vol}(\{1\}) = 3,$$

$$\mathrm{vol}(\{2, 3, 4\}) = 4 + 4 + 3 = 11.$$

Therefore:

$$\mathrm{Ncut}(\mathcal{P}_2) = \frac{3}{3} + \frac{3}{11}.$$

Hence:

$$\boxed{\mathrm{Ncut}(\mathcal{P}_2) = \frac{14}{11} \approx 1.273}.$$

Solution 3: Laplacian energy

4. Use the partition indicator:

$$\boldsymbol{f} = [1, 1, 0, 0]^\top.$$

First:

$$\boldsymbol{L}\boldsymbol{f} = [0, 1, -1, 0]^\top.$$

Then:

$$\boldsymbol{f}^\top \boldsymbol{L}\boldsymbol{f} = [1, 1, 0, 0][0, 1, -1, 0]^\top = \boxed{1}.$$

This equals:

$$\mathrm{cut}(\mathcal{P}_1) = 1.$$

The Laplacian energy therefore measures the similarity weight crossing this binary partition.

Solution 3: preferred partition

5. Compare the normalized-cut values:

$$\mathrm{Ncut}(\mathcal{P}_1) \approx 0.286,$$

$$\mathrm{Ncut}(\mathcal{P}_2) \approx 1.273.$$

Since smaller normalized cut is preferred:

$$\boxed{\mathcal{P}_1 = \{1, 2\} \mid \{3, 4\}}.$$

This partition separates the graph across the weak edge of weight 1, while preserving the two strong edges of weight 3 inside the groups.

Tutorial 5 complete

Questions and discussion